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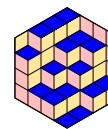


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ABSTRACT Motivated by higher homological algebra, we associate quivers to triangulations of even-dimensional cyclic polytopes and prove two results showing what information about the triangulation is encoded in the quiver. We first show that the cut quivers of Iyama and Oppermann correspond precisely to $2d$ -dimensional triangulations without interior $(d+1)$ -simplices. This implies that these triangulations form a connected subgraph of the flip graph. Our second result shows how the quiver of a triangulation can be used to identify mutable internal d -simplices. This points towards what a theory of higher-dimensional quiver mutation might look like and gives a new way of understanding flips of triangulations of even-dimensional cyclic polytopes.

In this paper we investigate the combinatorics of quivers associated to triangulations of even-dimensional cyclic polytopes. As shown by Oppermann and Thomas [30], such quivers provide the prototype for higher-dimensional cluster theory. This paper is written with two audiences in mind. On the one hand, it is mindful of polytopal combinatorialists who are interested in the structure of triangulations of cyclic polytopes. On the other hand, it may interest representation theorists or cluster theorists who are curious about the possibility of higher-dimensional cluster theory.

Since their introduction [13], cluster algebras have generated substantial amounts of fruitful research touching many areas of mathematics, including dynamical systems [14, 6, 33], Poisson geometry [19, 20], and Teichmüller theory [10, 11]. A cluster algebra can be given by choosing a quiver with a variable assigned to each vertex, and subsequently generating new quivers by a process of mutation, with new variables given from old variables via “exchange relations”. A remarkable result is that the cluster algebras of finite cluster type are precisely those coming from Dynkin diagrams, just as in the Cartan–Killing classification [15]. Two particular topics which are connected with cluster algebras are surfaces [12] and representation theory of algebras [5, 4]. Surfaces can be used to produce cluster algebras via triangulations, whilst in representation theory, cluster algebras are categorified via cluster categories.

A particularly simple example of a cluster algebra coming from a surface is the cluster algebra of type A_n , where the clusters are in bijection with triangulations of a convex $(n+3)$ -gon [13]. Mutation of clusters corresponds to flipping a diagonal inside a quadrilateral. The quiver of a cluster can be easily constructed from the triangulation by drawing arrows between neighbouring arcs. In the type A cluster category, these quivers are the Gabriel quivers of the endomorphism algebras of the corresponding cluster-tilting objects.

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KEYWORDS. Cyclic polytopes, triangulations, quivers, mutation.

An active area of research within representation theory is higher Auslander–Reiten theory, which was introduced by Iyama in a series of papers [24, 23, 25]. This subject studies subcategories of module categories which behave like higher-dimensional versions of abelian categories, and shows that several classical results can be generalised to them. Higher Auslander–Reiten theory has found beautiful connections with other areas of mathematics, such as symplectic geometry [9], non-commutative algebraic geometry [22], and combinatorics [30, 35].

Cluster theory and higher Auslander–Reiten theory were connected in [30], where higher-dimensional cluster phenomena were discovered for type A . Here higher-dimensional cluster categories were defined, with the cluster-tilting objects corresponding to triangulations of even-dimensional cyclic polytopes. Mutation of clusters now corresponds to bistellar flips—the higher-dimensional analogue of flipping a polygon inside a quadrilateral.

Higher-dimensional cluster theory remains poorly understood. A higher cluster algebra has yet to be defined—if such a definition is indeed possible. It would be remarkable if cluster algebras were the two-dimensional instance of a more general phenomenon. The necessary ingredients for a higher cluster algebra would be a rule for quiver mutation and an exchange relation to produce new cluster variables after mutation. Whilst higher tropical exchange relations were exhibited in [30], it is known that naïvely detropicalising these relations does not work.

Triangulations of even-dimensional cyclic polytopes present themselves as the guide for how the higher-dimensional quiver combinatorics ought to work. Just as in the classical type A case, the quiver of a cluster arises both from the endomorphism algebra of the corresponding cluster-tilting object, and may also be constructed from the corresponding triangulation: the vertices of the quiver correspond to the internal d -simplices of the triangulation, which we refer to as d -arcs, with arrows between the d -arcs that are closest to each other.

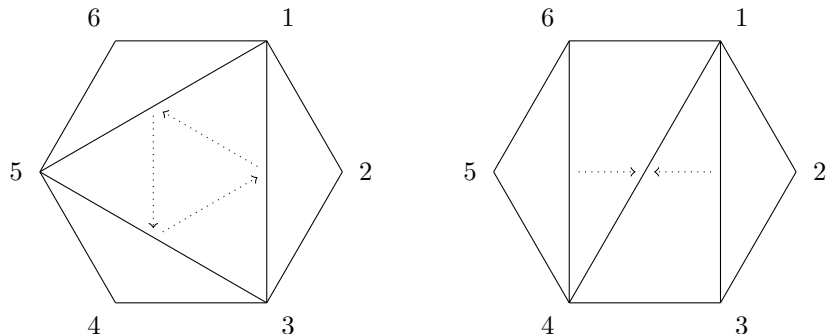
We investigate the information encoded by the quiver associated to a triangulation of a $2d$ -dimensional cyclic polytope. The best-understood quivers are those known as *cut quivers*, which were introduced in [26]. These quivers have a rule for mutation at sinks and sources [26]. Our first result shows that cut quivers correspond precisely to triangulations with no interior $(d + 1)$ -simplices and that these are also exactly the triangulations whose quivers are acyclic.

THEOREM 0.1 (Theorem 2.8). *A triangulation of a $2d$ -dimensional cyclic polytope has no interior $(d + 1)$ -simplices if and only if its quiver is acyclic, in which case its quiver is a cut quiver of type A .*

For $d = 1$, these cut quivers are precisely orientations of the A_n Dynkin diagram. Hence, this is a higher-dimensional generalisation of the fact that a triangulation of a convex polygon has no internal triangles if and only if its quiver is acyclic, and in this case the quiver is an orientation of the A_n Dynkin diagram, and thus provides a neat combinatorial description of a certain class of triangulations of an even-dimensional cyclic polytope. An application of this result is that the set of triangulations of a $2d$ -dimensional cyclic polytope without internal $(d + 1)$ -simplices forms a connected subgraph of the flip graph of the polytope.

Unlike for the 2-dimensional case, for $d > 1$ it is not possible to perform a bistellar flip at every internal d -simplex of a $2d$ -dimensional triangulation, or, equivalently, at every vertex of its quiver. This is an important difference with classical cluster theory, where a key property is that one can mutate a given cluster at every vertex of its quiver. This feature makes higher-dimensional cluster theory much more difficult to work with. Our second result uses the quiver of a triangulation to give a combinatorial criterion for identifying which d -simplices are *mutable*—that is, admit a bistellar flip.

FIGURE 1. Triangulations of polygons with and without interior triangles



We show how the arrows in the quiver can be partitioned into paths which we call *maximal retrograde paths*, and prove the following theorem.

THEOREM 0.2 (Theorem 3.11). *Let $\mathcal{T}$ be a triangulation of $C(n + 2d + 1, 2d)$. A d -arc of $\mathcal{T}$ is mutable if and only if it is not in the middle of a maximal retrograde path.*

This theorem gives a quiver-theoretic criterion for mutability, and hence points towards what a theory of higher-dimensional quiver mutation [15] could look like. Other extensions of quiver mutation have been of interest in the literature, such as to ice quivers [31]. As an application of this theorem, we give a rule for mutating cut quivers at vertices which are not necessarily sinks or sources. From the perspective of polyhedral geometry, this provides a visual way of understanding mutability for higher-dimensional triangulations, and makes it easier to compute bistellar flips of higher-dimensional triangulations by hand. In the case of polygon triangulations, all retrograde paths are of length one, so that the criterion imposes no restriction and all arcs are mutable.

Our work can also be seen in the context of combinatorial descriptions of algebraic mutation. For example, [29] gives a combinatorial description of the silting mutation from [1], whilst [17] gives a combinatorial procedure for tilting mutation. The algebraic objects corresponding to our triangulations and quivers are cluster-tilting objects in the $(d + 2)$ -angulated cluster categories of type A from [30]. Theorem 0.2 gives a combinatorial way of identifying the mutable summands of these cluster-tilting objects.

The structure of this paper is as follows. In Section 1 we give background to the paper, predominantly on triangulations of even-dimensional cyclic polytopes. In Section 2 we consider triangulations of $2d$ -dimensional cyclic polytopes without interior $(d + 1)$ -simplices and prove Theorem 0.1. In Section 3 we study mutability of d -simplices in terms of quivers and prove Theorem 0.2.

1. BACKGROUND

1.1. CONVENTIONS. We use $[m]$ to denote the set $\{1, \dots, m\}$. By $\binom{[m]}{k}$ we mean the set of subsets of $[m]$ of size k . For $l \in [m]$, using the notation of [28], we use $<_l$ to denote the cyclically shifted order on $[m]$ given by

$$l <_l l + 1 <_l \dots <_l m - 1 <_l m <_l 1 <_l \dots <_l l - 1.$$

For $r \geq 3$, $a_1 < \dots < a_r$ is a *cyclic ordering* if there is an $l \in [n + 2d + 1]$ such that $a_1 <_l \dots <_l a_r$.

Throughout this paper, we will often need to change the ordering on $[m]$ to $<_l$ for some l . We will usually do this tacitly, rather than explicitly writing $<_l$ in place of $<$. Similarly, we will usually be tacitly using arithmetic modulo $n + 2d + 1$, where this will be the number of vertices of our cyclic polytope. Hence, we shall write things like $j = i - 1$, when we mean $j \equiv i - 1 \pmod{n + 2d + 1}$. Our convention here will also be that our equivalence-class representatives for arithmetic modulo $n + 2d + 1$ will be $[n + 2d + 1]$ rather than $\{0, 1, \dots, n + 2d\}$. This is because we shall label the vertices of our cyclic polytope by $[n + 2d + 1]$.

Finally, given a tuple $A \in [m]^{k+1}$, we will denote the elements of A by $(a_0, a_1, \dots, a_k)$ where $a_0 <_l a_1 <_l \dots <_l a_k$ in a cyclically shifted order to be determined by the context. By default this will be the usual order on $[m]$. Because we wish sometimes to change this order, we are also tacitly identifying tuples which are the same up to cyclically shifted reordering. The same applies to other letters of the alphabet: the upper case letter denotes the tuple; the lower case letter is used for the entries, which are ordered according to their index, which starts from 0. For example, if we write $A = (1, 3, 5)$, then we have $a_0 = 1$, $a_1 = 3$, and $a_2 = 5$. If we changed the cyclic ordering from $<_1$ to $<_3$, then we would have $A = (3, 5, 1)$ with $a_0 = 3$, $a_1 = 5$, and $a_2 = 1$.

1.2. TRIANGULATIONS OF CYCLIC POLYTOPES. Cyclic polytopes should be thought of as higher-dimensional analogues of convex polygons. General introductions to this class of polytopes can be found in [36, Lecture 0], [21, 4.7], [7, Section 6.1] and [2, Chapter VI]. Let $n \geq 0$ and $\delta \geq 1$ and consider the curve defined by $p_t = (t, t^2, \dots, t^\delta) \in \mathbb{R}^\delta$ for $t \in \mathbb{R}$, which is known as the *moment curve*. Choose $t_1, t_2, \dots, t_{n+\delta+1} \in \mathbb{R}$ such that $t_1 < t_2 < \dots < t_{n+\delta+1}$. The convex hull $\text{conv}\{p_{t_1}, p_{t_2}, \dots, p_{t_{n+\delta+1}}\}$ is a *cyclic polytope* $C(n + \delta + 1, \delta)$. A (*geometric*) *triangulation* of a cyclic polytope $C(n + \delta + 1, \delta)$ is a collection of (geometric) δ -simplices whose interiors are pairwise disjoint and whose union is $C(n + \delta + 1, \delta)$.

A *facet* of $C(n + \delta + 1, \delta)$ is a face of codimension one. A *circuit* of a cyclic polytope $C(n + \delta + 1, \delta)$ is a pair, (Z_+, Z_-) , of sets of vertices of $C(n + \delta + 1, \delta)$ which are inclusion-minimal with respect to the property $\text{conv}(Z_+) \cap \text{conv}(Z_-) \neq \emptyset$, in the sense that if $Z'_+ \subseteq Z_+$ and $Z'_- \subseteq Z_-$ with $\text{conv}(Z'_+) \cap \text{conv}(Z'_-) \neq \emptyset$, then $Z'_+ = Z_+$ and $Z'_- = Z_-$.

The circuits and facets of a cyclic polytope are independent of its particular geometric realisation given by the choice of points on the moment curve, by [18, 3]. This implies that whether or not a collection of ordered $(\delta + 1)$ -tuples of $[n + \delta + 1]$ forms a triangulation of $C(n + \delta + 1, \delta)$ is likewise independent of the particular geometric realisation. Hence, in this paper we primarily consider triangulations combinatorially: a *combinatorial triangulation* of $C(n + \delta + 1, \delta)$ is a collection of ordered $(\delta + 1)$ -tuples with entries in $[n + \delta + 1]$ which gives a geometric triangulation of $C(n + \delta + 1, \delta)$. Unless otherwise specified, when we write ‘triangulation’ we mean a combinatorial triangulation. Likewise, if we say that A is a k -simplex, we mean that A is an ordered $(k + 1)$ -tuple with entries in $[n + \delta + 1]$. We write $|A|$ to denote the geometric realisation of A as a geometric simplex which is the convex hull of $k + 1$ points on the moment curve.

In this paper we are exclusively interested in triangulations of even-dimensional cyclic polytopes. These were given an elegant combinatorial description in [30], which we now explain. This combinatorial description was in turn used to connect triangulations of even-dimensional cyclic polytopes to representation theory of algebras. We will sometimes comment on these connections, but this paper does not require the reader to be familiar with representation theory.

A d -simplex of a triangulation of $C(n + 2d + 1, 2d)$ is *internal* if it does not lie within a facet of $C(n + 2d + 1, 2d)$. It is clear that a triangulation of a convex polygon is determined by the arcs of the triangulation; similarly, a triangulation of $C(n + 2d + 1, 2d)$ is determined by the internal d -simplices of the triangulation, by a theorem of Dey [8]. For brevity, we refer to an internal d -simplex of a triangulation $\mathcal{T}$ of $C(n + 2d + 1, 2d)$ as a d -arc of $\mathcal{T}$. We write $\mathbf{d}\text{-arcs}(\mathcal{T})$ for the set of d -arcs of $\mathcal{T}$. We say that a $(d + 1)$ -simplex of $\mathcal{T}$ is *interior* if all of its facets are d -arcs.

In order to recall the description of the triangulations of $C(n + 2d + 1, 2d)$ from [30], and for use later in the paper, we denote the following sets.

$$\tilde{\mathbf{I}}_{n+2d+1}^d = \left\{ (a_0, \dots, a_d) \in \mathbb{Z}^{d+1} \mid \begin{array}{l} \forall i \in \{0, 1, \dots, d-1\}, a_{i+1} \geq a_i + 2 \\ \text{and } a_d + 2 \leq a_0 + n + 2d + 1 \end{array} \right\};$$

$$\circlearrowleft \mathbf{I}_{n+2d+1}^d = \tilde{\mathbf{I}}_{n+2d+1}^d \cap [n + 2d + 1]^{d+1}.$$

REMARK 1.1. Algebraically, $\tilde{\mathbf{I}}_{n+2d+1}^d$ labels the d -cluster-tilting subcategory of the derived category of A_n^d , whilst $\circlearrowleft \mathbf{I}_{n+2d+1}^d$ labels its cluster category [30]. Here A_n^d is the higher Auslander algebra of type A ; see [25].

A d -simplex $A \in [n + 2d + 1]^{d+1}$ is a d -arc in $C(n + 2d + 1, 2d)$ if and only if $A \in \circlearrowleft \mathbf{I}_{n+2d+1}^d$. A d -simplex A and a d -simplex B are *intertwining* if

$$a_0 < b_0 < a_1 < b_1 < \dots < a_d < b_d$$

is a cyclic ordering, in which case we write $A \bowtie B$. Note that A and B being intertwining is therefore independent of the cyclically shifted order $<_l$ we may choose on $[n + 2d + 1]$. A collection of d -simplices is called *non-intertwining* if no pair of its elements are intertwining. The circuits of the cyclic polytope $C(n + 2d + 1, 2d)$ are pairs of intertwining d -arcs.

THEOREM 1.2 ([30, Theorem 2.3, Theorem 2.4]). *There is a bijection between elements of $\circlearrowleft \mathbf{I}_{n+2d+1}^d$ and d -arcs of $C(n + 2d + 1, 2d)$, which induces a bijection between non-intertwining collections of $\binom{n+d-1}{d}$ d -simplices in $\circlearrowleft \mathbf{I}_{n+2d+1}^d$ and triangulations of $C(n + 2d + 1, 2d)$, via sending*

$$\mathcal{T} \mapsto \mathbf{d}\text{-arcs}(\mathcal{T}).$$

Triangulations of cyclic polytopes can be *mutated* by operations known as *bistellar flips* [7, Section 2.4]. The following theorem can be taken as a definition of this operation in the case of triangulations of $C(n + 2d + 1, 2d)$.

THEOREM 1.3 ([30, Theorem 4.1]). *Triangulations $\mathcal{T}$ and $\mathcal{T}'$ of $C(n + 2d + 1, 2d)$ are bistellar flips of each other if and only if $\mathbf{d}\text{-arcs}(\mathcal{T})$ and $\mathbf{d}\text{-arcs}(\mathcal{T}')$ have all but one element in common.*

1.3. CUTS AND SLICES. We will associate quivers to triangulations of even-dimensional cyclic polytopes and show what properties of the triangulation are encoded in the quiver. Indeed, we now define the quivers which are higher analogues of orientations of the A_n Dynkin diagram, following [26]. The main result of the first part of this paper is that these quivers characterise triangulations of $C(n + 2d + 1, 2d)$ with no internal $(d + 1)$ -simplices.

Let $Q^{(d,n)}$ be the quiver with vertices

$$Q_0^{(d,n)} := \left\{ (a_0, \dots, a_d) \in \mathbb{Z}_{\geq 0}^{d+1} \mid \sum_{i=0}^d a_i = n - 1 \right\}$$

FIGURE 2. Examples of the quivers $Q^{(d,n)}$

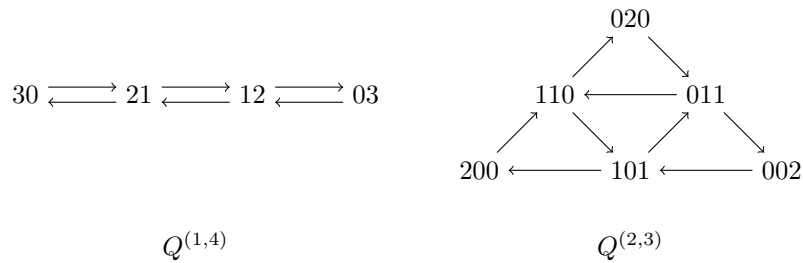
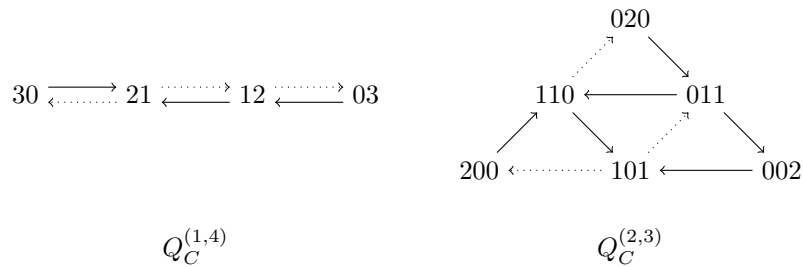


FIGURE 3. Cuts of the quivers $Q^{(d,n)}$



and arrows

$$Q_1^{(d,n)} := \left\{ A \rightarrow A + F_i \mid A, A + F_i \in Q_0^{(d,n)} \right\},$$

where

$$F_i = (\dots, 0, -1, \overset{i}{1}, \overset{i+1}{0}, \dots),$$

with

$$F_d = (1, 0, \dots, 0, -1).$$

We say that arrows $A \rightarrow A + F_i$ are *arrows of type i* . See Figure 2 for pictures of these quivers. A subset $C \subseteq Q_1^{(d,n)}$ is called *cut* if it contains exactly one arrow from each $(d+1)$ -cycle in $Q^{(d,n)}$. Given a cut C , we write $Q_C^{(d,n)}$ for the quiver with arrows $Q_1^{(d,n)} \setminus C$ and refer to this as the *cut quiver*. Examples of cut quivers can be seen in Figure 3. Note that the cut quivers of $Q^{(1,n)}$ are precisely the orientations of the A_n Dynkin diagram. Hence, for $d > 1$, we think of cut quivers of $Q^{(d,n)}$ as higher analogues of orientations of the A_n Dynkin diagram.

Iyama and Oppermann [26] show that cut quivers of $Q^{(d,n)}$ are precisely the quivers than can be realised as *slices* of another family of quivers, denoted $\tilde{Q}^{(d,n)}$, which we now define. Let $\tilde{Q}^{(d,n)}$ be the quiver with vertices

$$\tilde{Q}_0^{(d,n)} := \tilde{\mathbf{I}}_{n+2d+1}^d$$

and arrows

$$\tilde{Q}_1^{(d,n)} := \left\{ A \rightarrow A + E_i \mid A, A + E_i \in \tilde{Q}_0^{(d,n)} \right\},$$

where

$$E_i := (\dots, 0, \overset{i}{1}, 0, \dots).$$

FIGURE 4. $\tilde{Q}^{(1,3)}$, with a slice shown in blue

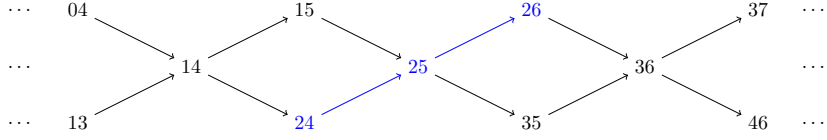
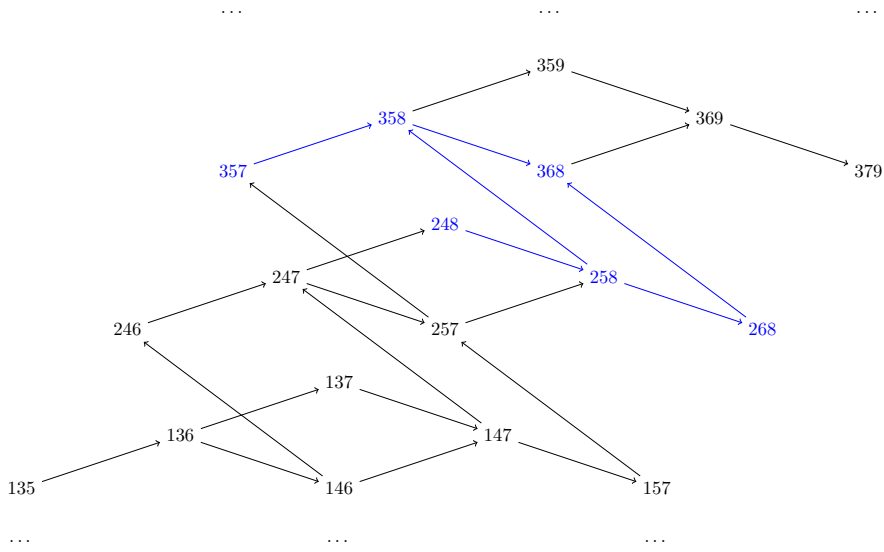


FIGURE 5. $\tilde{Q}^{(2,3)}$, with a slice shown in blue



Given two $(d+1)$ -tuples $A, B \in \mathbb{Z}^{d+1}$, we write $A \leq B$ if $a_i \leq b_i$ for all $i \in \{0, 1, \dots, d\}$. We use $A < B$ analogously. We write $A \rightsquigarrow B$ if there is a path from A to B . Note that, given $A, B \in \tilde{Q}_0^{(d,n)}$, there is a path $A \rightsquigarrow B$ in $\tilde{Q}^{(d,n)}$ if and only if $A \leq B$.

We denote by ν_d the automorphism of $\tilde{Q}^{(d,n)}$ given by $A \mapsto A - \mathbf{1}$. Here, $A - \mathbf{1} := (a_0 - 1, \dots, a_d - 1)$. We also write $A + \mathbf{1}$, in an analogous way.

DEFINITION 1.4. We denote by $\pi: \tilde{Q}_0^{(d,n)} \rightarrow \circlearrowleft \mathbf{I}_{n+2d+1}^d$ the map given by

$$A \mapsto \text{sort}(\pi(a_0), \pi(a_1), \dots, \pi(a_d)),$$

where $\pi(a_i) := a_i \pmod{n+2d+1}$ and sort indicates that we should sort the tuple so that it is increasing with respect to the usual order on $[n+2d+1]$.

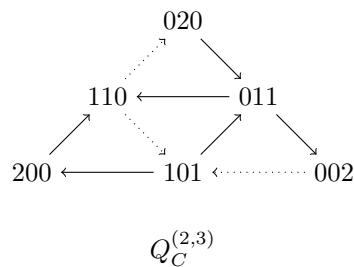
Note that the definition of $\tilde{\mathbf{I}}_{n+2d+1}^d$ guarantees that $\pi(A)$ does not contain any repeated entries.

Following [26, Definition 5.20], we define a *slice* of $\tilde{Q}^{(d,n)}$ to be a full subquiver S of $\tilde{Q}^{(d,n)}$ such that:

- (1) Any ν_d orbit in $\tilde{Q}^{(d,n)}$ contains precisely one vertex which belongs to S .
- (2) S is convex, i.e., for any path P in $\tilde{Q}^{(d,n)}$ connecting two vertices in S , all vertices appearing in P belong to S .

Slices are shown in blue in Figure 4 and Figure 5.

FIGURE 6. Mutating the cut of $Q^{(2,3)}$ from Figure 3 at vertex 101



Given a slice S of $\tilde{Q}^{(d,n)}$, one can find a cut C_S of $Q^{(d,n)}$ such that $Q_{C_S}^{(d,n)}$ is isomorphic to S with the arrows F_i of $Q_{C_S}^{(d,n)}$ corresponding to arrows $A \rightarrow A + E_i$ in S [26, Theorem 5.24].

REMARK 1.5. These combinatorial constructions of course have representation-theoretic interpretations, which are explained in [26]. Cuts and slices correspond to iterated d -APR tilts. The map ν_d is the d -th desuspension of the Serre functor on the derived category. The map π corresponds to the projection from the derived category to the $(d + 2)$ -angulated cluster category [30].

1.3.1. *Mutation of cuts and slices.* Cuts and slices can be mutated, as was defined in [26]. This corresponds to algebraic mutation and also to bistellar flips.

- Let C be a cut of $Q^{(d,n)}$ and let x be a source of $Q_C^{(d,n)}$. Define a subset $\mu_x^+(C)$ of $Q_1^{(d,n)}$ by removing all arrows in C which end at x and adding all arrows in $Q_1^{(d,n)}$ which begin at x . Dually, if x is a sink of $Q_C^{(d,n)}$, define $\mu_x^-(C)$ by removing all arrows in C which begin at x and adding all arrows in $Q_1^{(d,n)}$ which end at x . By [26, Proposition 5.14], we have that $\mu_x^+(C)$ and $\mu_x^-(C)$ are also cuts of $Q^{(d,n)}$. This process is illustrated in Figure 6.
- Let S be a slice of $\tilde{Q}^{(d,n)}$. If x is a source of S , then define a full subquiver $\mu_x^+(S)$ of $\tilde{Q}^{(d,n)}$ by removing x from S and adding $\nu_d^{-1}x$ [26, Definition 5.25]. Dually, if x is a sink of S , define a full subquiver $\mu_x^-(S)$ by removing x and adding ν_dx . This process is illustrated in Figure 7.

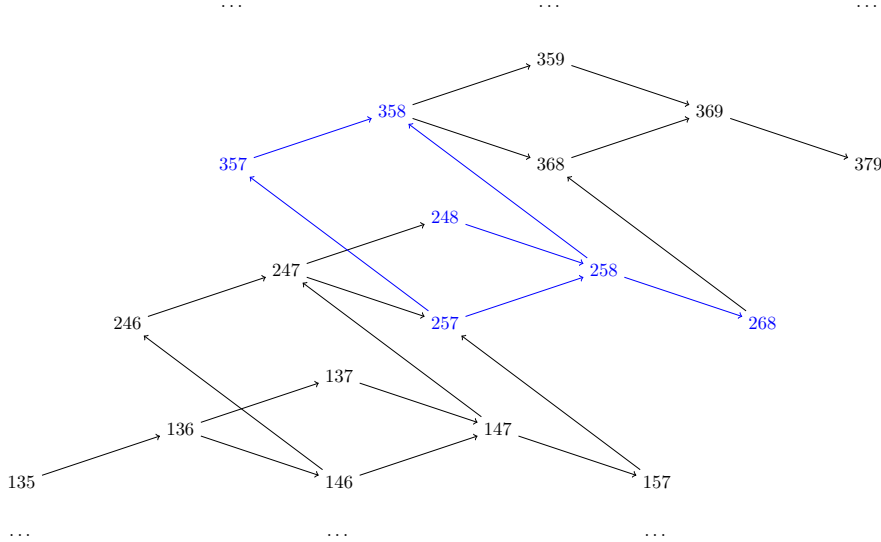
If C_S is the cut corresponding to a slice S then $C_{\mu_x^+S} = \mu_x^+(C_S)$ and $C_{\mu_x^-S} = \mu_x^-(C_S)$, provided x is a source or sink, respectively. Here we abuse notation by using x to refer both to the relevant vertex of S and to the relevant vertex of $Q_{C_S}^{(d,n)}$.

2. TRIANGULATIONS WITHOUT INTERIOR $(d + 1)$ -SIMPLICES

In this section we prove our combinatorial description of triangulations of $C(n + 2d + 1, 2d)$ without interior $(d + 1)$ -simplices and use this description to show that this class of triangulations is connected by bistellar flips. From the perspective of polyhedral geometry, this provides a combinatorial description of the triangulations without interior $(d + 1)$ and shows how they sit within the class of all triangulations. From the perspective of representation theory, this gives a geometric description of iterated d -APR tilts of A_n^d .

2.1. THE QUIVER OF A TRIANGULATION. We first define the quiver of a triangulation, which is the higher-dimensional version of the quiver of a polygon triangulation—see, for instance, [16, Section 3] and [34, Definition 2.12].

FIGURE 7. Mutating the slice from Figure 5 at the vertex 368



DEFINITION 2.1. Given a triangulation $\mathcal{T}$ of $C(n + 2d + 1, 2d)$, we define the quiver $Q(\mathcal{T})$ of $\mathcal{T}$ to be the following directed graph. The vertices are

$$Q_0(\mathcal{T}) = \text{d-arcs}(\mathcal{T}).$$

The arrows are given by $A \rightarrow B$ with $A \neq B$ such that $(A - \mathbf{1}) \circ B$ and there exists no $A' \in \text{d-arcs}(\mathcal{T}) \setminus \{A, B\}$ with $(A - \mathbf{1}) \circ A'$ and $(A' - \mathbf{1}) \circ B$. Recall our convention of taking $A - \mathbf{1}$ modulo $n + 2d + 1$.

We define the quiver in this way so that it coincides with the Gabriel quiver of the endomorphism algebra of the cluster-tilting object corresponding to the triangulation. The arrows mirror the description of the homomorphisms in the cluster category $\mathcal{O}_{A_n^d}$ from [30]. However, we now prove a simpler description of the quiver, given in Corollary 2.4. In order to obtain this, we first make some observations about cyclic polytopes. Changing the ordering on $[n + 2d + 1]$ from $<_1$ to $<_l$ induces a (combinatorial) automorphism of the cyclic polytope $C(n + 2d + 1, 2d)$; see [27]. We think of this automorphism as reorienting the cyclic polytope. Generalising [30], given a $2d$ -simplex T of $C(n + 2d + 1, 2d)$ and an ordering $<_l$ of $[n + 2d + 1]$ under which T is ordered $T = (t_0, t_1, \dots, t_{2d})$, we write $e_l(T) = (t_0, t_2, \dots, t_{2d-2}, t_{2d})$. If the ordering is determined by the context, we simply write $e(T)$.

LEMMA 2.2. Given an ordering $<_l$ of $[n + 2d + 1]$ and a d -arc A of a triangulation $\mathcal{T}$ of $C(n + 2d + 1, 2d)$, there is a $2d$ -simplex T of $\mathcal{T}$ such that $e_l(T) = A$.

Proof. This follows from applying [30, Proposition 2.13] in the orientation given by $<_l$. $\square$

PROPOSITION 2.3. Suppose that $A \rightarrow B$ is an arrow in $Q(\mathcal{T})$. Then A and B share all but one entry, and there is a $2d$ -simplex T of $\mathcal{T}$ such that A and B are both faces of T .

Proof. We have that $(A - \mathbf{1}) \circ B$. It must be the case that A and B have at least one common entry, otherwise $A \circ B$. Hence, by re-ordering, we may assume that $a_0 = b_0$

and $a_{d-1} < a_d < b_d$. There must be a $2d$ -simplex T of $\mathcal{T}$ such that $e(T) = B$. We label the vertices of T by $T = (b_0, x_1, b_1, x_2, \dots, b_{d-1}, x_d, b_d)$.

We cannot have that $b_{i-1} < x_i \leq a_i - 1$ for all $i \in [d]$, otherwise $A \bowtie (x_1, x_2, \dots, x_d, b_d)$, which is impossible since A and $(x_1, x_2, \dots, x_d, b_d)$ are both d -arcs of $\mathcal{T}$. Hence there is an $i \in [d]$ such that $b_{i-1} < a_i - 1 < x_i$. Then $(b_0, b_1, \dots, b_{i-1}, x_i, b_{i+1}, b_{i+2}, \dots, b_d)$ is a d -arc of $\mathcal{T}$, since $b_{i-1} < a_i - 1 < x_i$ and $x_i < b_i < b_{i+1}$. Moreover, $(A - \mathbf{1}) \bowtie (b_0, b_1, \dots, b_{i-1}, x_i, b_{i+1}, b_{i+2}, \dots, b_d)$ and $(b_0 - 1, \dots, b_{i-1} - 1, x_i - 1, b_{i+1} - 1, \dots, b_d - 1) \bowtie B$. Since $A \rightarrow B$ is an arrow in $Q(\mathcal{T})$, we must have $A = (b_0, b_1, \dots, b_{i-1}, x_i, b_{i+1}, b_{i+2}, \dots, b_d)$. (In fact, by the ordering we have chosen, we must have that $i = d$.) Therefore A and B are both faces of the $2d$ -simplex T and they share all but one entry, as desired. $\square$

COROLLARY 2.4. *The arrows of $Q(\mathcal{T})$ are*

$$Q_1(\mathcal{T}) = \left\{ A \rightarrow A + rE_i \mid \begin{array}{l} A, A + rE_i \in \circlearrowleft \mathbf{I}_{n+2d+1}^d, \\ \nexists s \in [r-1] \text{ such that } A + sE_i \in Q_0(\mathcal{T}) \end{array} \right\}.$$

We call arrows $A \rightarrow A + rE_i$ *arrows of type i* , analogously to our terminology of arrows of type i in $Q^{(d,n)}$ from Section 1.3. Note that this labelling depends on the choice of an ordering $<_l$.

For an arrow α , we denote the *head* $h(\alpha)$ and the *tail* $t(\alpha)$ such that $t(\alpha) \xrightarrow{\alpha} h(\alpha)$. We say that α is *incident* at $t(\alpha)$ and $h(\alpha)$. Given a quiver Q with vertices $A, B \in Q_0$, by a *path* in Q from A to B we mean a finite sequence of arrows $\alpha_1 \dots \alpha_r$ such that $t(\alpha_1) = A, h(\alpha_r) = B$ and $h(\alpha_{i-1}) = t(\alpha_i)$ for all $i \in \{2, 3, \dots, r\}$. If there is a path from A to B , then we write $A \rightsquigarrow B$. We note the following property concerning paths in $Q(\mathcal{T})$ which will be useful later.

LEMMA 2.5. *Given a triangulation $\mathcal{T}$ of $C(n + 2d + 1, 2d)$ and $A, B \in Q_0(\mathcal{T})$ with $(A - \mathbf{1}) \bowtie B$, there is a path from A to B in $Q(\mathcal{T})$.*

Proof. This is clear from Definition 2.1, using induction. $\square$

2.2. QUIVER DESCRIPTION. With these preliminaries taken care of, we now move to prove the first main result of this paper, which describes triangulations $\mathcal{T}$ without interior $(d + 1)$ -simplices in terms of their quivers $Q(\mathcal{T})$. Recall that a $(d + 1)$ -simplex of a triangulation $\mathcal{T}$ of $C(n + 2d + 1, 2d)$ is interior if all of its facets are d -arcs.

Slices of $\tilde{Q}^{(d,n)}$ give triangulations of $C(n + 2d + 1, 2d)$. We give a direct combinatorial proof of this, although it can also be deduced from the fact that slices correspond to iterated d -APR tilts [26, Theorem 4.15], which implies that projecting to the cluster category will give a triangulation [30, Theorem 6.4]. Recall the map π from Definition 1.4.

PROPOSITION 2.6. *If S is a slice of $\tilde{Q}^{(d,n)}$, then the vertices $\pi(S_0)$ give a triangulation of $C(n + 2d + 1, 2d)$.*

Proof. There are as many ν_d -orbits as there are elements of $\circlearrowleft \mathbf{I}_{n+2d+1}^d$ containing 1, namely $\binom{n+d-1}{d}$. Suppose that there exist $\pi(A)$ and $\pi(B)$ in $\pi(S_0)$ with $\pi(A) \bowtie \pi(B)$. We assume without loss of generality that $a_0 < b_0$, noting that $\pi(A) \bowtie \pi(B)$ implies that $a_i \neq b_i$ for all i . We claim that $A < B$. Suppose for contradiction that $b_i < a_i$ for some i . We may choose the minimal i such that this is the case. Then $a_{i-1} < b_{i-1} < b_i < a_i$. Since we must have

$$\pi(a_{i-1}) < \pi(b_{i-1}) < \pi(a_i) < \pi(b_i),$$

we must have $a_i - a_{i-1} > n + 2d + 1$. Hence $a_d - a_0 > n + 2d + 1 > n + 2d - 1$, which contradicts $A \in \tilde{\mathbf{I}}_{n+2d+1}^d$.

Therefore $A < A + 1 \leq B$, which means that here is then a path $A \rightsquigarrow A + 1 \rightsquigarrow B$, so $A + 1 \in \pi(S_0)$ by convexity. But this contradicts the fact that S contains one vertex from every ν_d -orbit. Hence $\pi(S_0)$ is a non-intertwining subset of ${}^{\circ}\mathbf{I}_{n+2d+1}^d$ of size $\binom{n+d-1}{d}$, and so gives a triangulation of $C(n + 2d + 1, 2d)$. $\square$

A similar argument also shows the following lemma, which will be useful later.

LEMMA 2.7. *If S is a convex subquiver of $\tilde{Q}^{(d,n)}$ such that $\pi(S_0)$ is a triangulation of $C(n + 2d + 1, 2d)$, then S is a slice.*

Proof. Suppose that S is a convex subquiver such that $\pi(S_0)$ is a triangulation of $C(n + 2d + 1, 2d)$. Suppose for contradiction that S possesses two vertices A and B which are in the same ν_d -orbit. There is then either a path $A \rightsquigarrow B$ or a path $B \rightsquigarrow A$. Without loss of generality, we suppose the former. But then there is a path $A \rightsquigarrow A + 1 \rightsquigarrow B$, so we must have $A + 1 \in S_0$ by convexity. This is a contradiction, since $\pi(A)$ and $\pi(A + 1)$ are intertwining. $\square$

The main theorem of this section is as follows. This shows how the quivers of triangulations detect the structure of the triangulation, and gives a geometric description of iterated d -APR tilts of A_n^d .

THEOREM 2.8. *A triangulation $\mathcal{T}$ of $C(n + 2d + 1, 2d)$ has no interior $(d + 1)$ -simplices if and only if its quiver is a cut of $Q^{(d,n)}$, and this is the case if and only if its quiver has no cycle.*

Our strategy for proving this theorem is to prove facts about the quivers $Q(\mathcal{T})$ for triangulations $\mathcal{T}$ of $C(n + 2d + 1, 2d)$ without internal $(d + 1)$ -simplices. We then use these properties to show that one can realise $Q(\mathcal{T})$ as a slice of $\tilde{Q}^{(d,n)}$, which will imply that $Q(\mathcal{T})$ is a cut of $Q^{(d,n)}$. The idea of the proofs of both of the following lemmas is that if $Q(\mathcal{T})$ does not possess a certain property, then we can find an interior $(d + 1)$ -simplex in $\mathcal{T}$.

LEMMA 2.9. *If $\mathcal{T}$ has no interior $(d + 1)$ -simplices, then the arrows in $Q(\mathcal{T})$ are all of the form $A \rightarrow A + E_i$.*

Proof. By Proposition 2.3, every arrow is of the form $A \rightarrow A + rE_i$ for some $r > 0$ and for each such arrow, we have that $(a_0, a_1, \dots, a_i, a_i + r, a_{i+1}, a_{i+2}, \dots, a_d)$ is a face of a $2d$ -simplex of $\mathcal{T}$. If $r > 1$, then this is an interior $(d + 1)$ -simplex. $\square$

LEMMA 2.10. *Suppose that $\mathcal{T}$ has no interior $(d + 1)$ -simplices. If $A \rightarrow A + E_i \rightarrow A + E_i + E_j$ is a sequence of arrows in $Q(\mathcal{T})$, then, if $A + E_j \in {}^{\circ}\mathbf{I}_{n+2d+1}^d$, the sequence $A \rightarrow A + E_j \rightarrow A + E_i + E_j$ is also in $Q(\mathcal{T})$.*

Proof. We assume $A + E_j \in {}^{\circ}\mathbf{I}_{n+2d+1}^d$. By re-ordering, we may assume that $i = d$. Then there is a $2d$ -simplex T of $\mathcal{T}$ such that $e(T) = A + E_j + E_d$ by Lemma 2.2.

Let $T = (a_0, x_1, a_1, \dots, x_j, a_j + 1, x_{j+1}, \dots, x_d, a_d + 1)$. If $x_d = a_d$, then $A + E_j$ is a d -face of T and hence a d -arc of $\mathcal{T}$. Hence, suppose for contradiction that $a_d \neq x_d$, so that $a_{d-1} < x_d \leq a_d - 1$. Note that if $x_j = a_{j-1} + 1$, then $(a_0, a_1, \dots, a_d) \vee (x_1, x_2, \dots, x_d, a_d + 1)$, which is a d -face of T . Hence $a_{j-1} + 2 \leq x_j < a_j + 1$. Then $(a_0, a_1, \dots, a_{j-1}, x_j, x_{j+1}, \dots, x_d, a_d + 1)$ is a $(d + 1)$ -face of T and an interior $(d + 1)$ -simplex of $\mathcal{T}$, a contradiction. $\square$

We now prove some facts about the quivers $\tilde{Q}^{(d,n)}$, which will be useful in proving the main theorem of this section. These will be used in combination with the previous two lemmas to show that one can realise $Q(\mathcal{T})$ as a slice if $\mathcal{T}$ has no interior $(d + 1)$ -simplices. We say that a full subquiver P of $\tilde{Q}^{(d,n)}$ is *switching-closed* if whenever

$A \rightarrow A + E_i \rightarrow A + E_i + E_j$ is a sequence of arrows in P and $A + E_j \in \tilde{Q}_0^{(d,n)}$, then the sequence of arrows $A \rightarrow A + E_j \rightarrow A + E_i + E_j$ is also in P .

LEMMA 2.11. *Let P be a switching-closed full subquiver of $\tilde{Q}^{(d,n)}$. If there is a path $A \rightsquigarrow B$ in P , all other paths $A \rightsquigarrow B$ in $\tilde{Q}^{(d,n)}$ must also lie in P .*

Proof. Suppose that we have a path $A \rightsquigarrow B$ in P . The length of any such path is $\sum_{i=0}^d (b_i - a_i)$. We prove the claim by induction on this quantity. The base case, where the length is 1, follows from the fact that P is a full subquiver of $\tilde{Q}^{(d,n)}$.

For the inductive step, we assume that the claim holds for all X and Y with $\sum_{i=0}^d (y_i - x_i) < \sum_{i=0}^d (b_i - a_i)$. We have that B is the head of up to $(d+1)$ arrows, namely the ones with tails $B - E_0, B - E_1, \dots, B - E_d$, provided these are vertices of $\tilde{Q}^{(d,n)}$. Suppose that $B - E_i$ is the penultimate vertex of our path $A \rightsquigarrow B$.

Choose a vertex $B - E_j \in \tilde{Q}_0^{(d,n)}$, where $i \neq j$. If $A \not\leq B - E_j$, then we may ignore this vertex, since there can be no paths from A to B through it. Hence we assume that $A \leq B - E_j$. This implies that $A \leq B - E_i - E_j \leq B - E_i$, so $B - E_i - E_j \in P_0$ by the induction hypothesis applied to A and $B - E_i$.

Since P is switching-closed, we have that $B - E_j \in P_0$, since $B - E_i - E_j, B - E_i, B \in P_0$. By the induction hypothesis, all paths $A \rightsquigarrow B - E_j$ lie in P , and hence all paths $A \rightsquigarrow B$ passing through $B - E_j$ lie in P . The result follows. $\square$

By a *walk* in a quiver Q from A to B we mean a finite sequence of arrows $\beta_1 \beta_2 \dots \beta_s$ such that β_1 is incident at A , β_s is incident at B , and β_{i-1} and β_i are incident at a common vertex for all $i \in \{2, 3, \dots, s\}$. In this case, we write $A \dashrightarrow B$. That is, a path only consists of forwards arrows, but a walk may contain backwards arrows as well.

LEMMA 2.12. *Let P be a connected switching-closed full subquiver of $\tilde{Q}^{(d,n)}$. If $A, B \in P_0$ are such that there is a path $A \rightsquigarrow B$ in $\tilde{Q}^{(d,n)}$, then there is a path $A \rightsquigarrow B$ in P .*

Proof. Let $A, B \in P_0$. Suppose that there is a path $A \rightsquigarrow B$ in $\tilde{Q}^{(d,n)}$. There is certainly a walk $W: A \dashrightarrow B$ in P , since P is connected. We prove that there is also a path $A \rightsquigarrow B$ in P by induction on the number of backwards arrows in this walk. The base case, in which there are zero backwards arrows in the walk, is immediate.

Hence we suppose for induction that the claim holds for walks with fewer backwards arrows than W . We may assume that the final arrow in W is a backwards one, otherwise we may remove the final arrow and consider instead the walk $A \dashrightarrow B'$, where $A \dashrightarrow B' \rightarrow B$ is the original walk W .

Therefore we can assume that our walk is of the form $A \dashrightarrow C \leftarrow B$, where $C \in P$. By the induction hypothesis, we can replace this with a walk of the form $A \rightsquigarrow C \leftarrow B$ in P . Moreover, by Lemma 2.11 we have that all paths $A \rightsquigarrow C$ in $\tilde{Q}^{(d,n)}$ lie in P . Since we have a path $A \rightsquigarrow B$ in $\tilde{Q}^{(d,n)}$, we have $A \leq B$ and, moreover, $A \leq B \leq C$. There is therefore a path $A \rightsquigarrow B \rightsquigarrow C$ in $\tilde{Q}^{(d,n)}$. This path is in P , since every path $A \rightsquigarrow C$ is in P , which gives the desired path $A \rightsquigarrow B$ in P . $\square$

These two lemmas imply the following corollary.

COROLLARY 2.13. *Let P be a connected switching-closed full subquiver of $\tilde{Q}^{(d,n)}$. Then P is convex in $\tilde{Q}^{(d,n)}$.*

Proof. Suppose that P is a connected switching-closed full subquiver of $\tilde{Q}^{(d,n)}$. Let $A, B \in P_0$ be such that there is a path $A \rightsquigarrow B$ in $\tilde{Q}^{(d,n)}$. By Lemma 2.12, there is a path $A \rightsquigarrow B$ in P . Then, by Lemma 2.11, we have that all paths $A \rightsquigarrow B$ lie in P , and so P is convex. $\square$

This corollary is useful because it is easier to check the property of being connected and switching-closed than the property of being convex. Given a full subquiver P of

$\tilde{Q}^{(d,n)}$, we write $\bar{P}$ for the smallest switching-closed subquiver containing P . The subquiver $\bar{P}$ is well-defined since the intersection of a set of switching-closed full subquivers is switching-closed, so $\bar{P}$ may be constructed as the intersection of all switching-closed subquivers containing P .

PROPOSITION 2.14. *A triangulation $\mathcal{T}$ contains an interior $(d+1)$ -simplex if and only if $Q(\mathcal{T})$ contains a cycle.*

Proof. We first suppose for contradiction that $\mathcal{T}$ contains no interior $(d+1)$ -simplices and that $Q(\mathcal{T})$ does contain a cycle. By Lemma 2.9, we can realise this cycle as a path $P: A \rightsquigarrow B$ in $\tilde{Q}^{(d,n)}$, where A is some vertex in the cycle in $Q(\mathcal{T})$ and $\pi(B) = A$ with $A \neq B$. We consider this path P as a subquiver of $\tilde{Q}^{(d,n)}$. By Lemma 2.11, every path $A \rightsquigarrow B$ in $\tilde{Q}^{(d,n)}$ must lie in $\bar{P}$. There is a path $A \rightsquigarrow A+1 \rightsquigarrow B$ in $\tilde{Q}^{(d,n)}$. Hence $A+1$ is a vertex of $\bar{P}$. By Lemma 2.10, if C is a vertex of $\bar{P}$, then $\pi(C)$ is a vertex of $Q(\mathcal{T})$. Therefore $\pi(A+1)$ is a vertex of $Q(\mathcal{T})$, but this is a contradiction, since $\pi(A+1)$ and A are intertwining.

We now suppose that $\mathcal{T}$ is a triangulation with an interior $(d+1)$ -simplex $(a_0, \dots, a_{d+1})$. Then $Q(\mathcal{T})$ has a cycle given by concatenating the paths

$$(a_0, a_1, \dots, a_{d-1}, a_d) \rightsquigarrow (a_0, a_1, \dots, a_{d-1}, a_{d+1}) \rightsquigarrow (a_0, a_1, \dots, a_{d-2}, a_d, a_{d+1}) \rightsquigarrow \dots \rightsquigarrow (a_1, a_2, \dots, a_d, a_{d+1}) \rightsquigarrow (a_0, a_1, \dots, a_d),$$

noting Lemma 2.5. □

REMARK 2.15. For $d = 1$, an interior triangle gives a 3-cycle in the quiver where all the vertices of the cycle are edges of the triangle. For $d > 1$, the cycle obtained in Proposition 2.14 may not exclusively have facets of the interior $(d+1)$ -simplex as its vertices. An example of this can be seen in Figure 9, where the interior 3-simplex is 1357, but the cycle is $135 \rightarrow 136 \rightarrow 137 \rightarrow 147 \rightarrow 157 \rightarrow 357 \rightarrow 135$. Here 136 and 147 are not faces of the interior 3-simplex 1357.

We are now ready to prove our first main result, noting that the second part of the statement has already been established by Proposition 2.14.

Proof of Theorem 2.8. First suppose that $\mathcal{T}$ has no interior $(d+1)$ -simplices. We consider the full subquiver R of $\tilde{Q}^{(d,n)}$ with vertices

$$R_0 = \left\{ B \in \tilde{Q}_0^{(d,n)} \mid \pi(B) \in Q_0(\mathcal{T}) \right\}.$$

We claim that this is disconnected and that each connected component gives $Q(\mathcal{T})$ by applying π . Let $A \in Q_0(\mathcal{T})$. If R is connected then it contains a walk $W: A \dashrightarrow (a_1, a_2, \dots, a_d, a_0+n+2d+1)$. We consider W as a subquiver of $\tilde{Q}^{(d,n)}$ and consider $\bar{W}$. By Lemma 2.12, $\bar{W}$ contains a path $P: A \rightsquigarrow (a_1, a_2, \dots, a_d, a_0+n+2d+1)$. By Lemma 2.10, if B is a vertex of $\bar{W}$, then $\pi(B)$ is a vertex of $Q(\mathcal{T})$. Hence all vertices of P give vertices of $Q(\mathcal{T})$, which therefore contains a cycle. But this contradicts Proposition 2.14.

By this argument, R is disconnected and, moreover, the vertices of each connected component are in bijection with the vertices of $Q(\mathcal{T})$ via π , since $Q(\mathcal{T})$ is connected. Moreover, the arrows in each connected component of R are the same as the arrows in $Q(\mathcal{T})$, by Lemma 2.9. Hence, by choosing one of the connected components, we obtain a full subquiver S of $\tilde{Q}^{(d,n)}$ such that $\pi(S) = Q(\mathcal{T})$. We then have that S is a switching-closed connected subquiver of $\tilde{Q}^{(d,n)}$, so S is convex by Lemma 2.13. Since $\pi(S_0)$ is a triangulation, it then follows from Lemma 2.7 that S is a slice. Hence $Q(\mathcal{T})$ is a cut of $Q^{(d,n)}$ by [26, Theorem 5.24].

Now suppose that $Q(\mathcal{T})$ is a cut of $Q^{(d,n)}$. Then $Q(\mathcal{T})$ cannot contain any cycles, since cut quivers can be realised as slices, which are full subquivers of $\tilde{Q}^{(d,n)}$, which does not contain any cycles. Then we obtain that $\mathcal{T}$ contains no interior $(d+1)$ -simplices by Proposition 2.14. $\square$

The *flip graph* of $C(n+2d+1, 2d)$ is the graph whose vertices are the triangulations of $C(n+2d+1, 2d)$, with edges connecting triangulations related by a bistellar flip. Theorem 2.8 implies that the set of triangulations of $C(n+2d+1, 2d)$ without interior $(d+1)$ -simplices is connected by bistellar flips.

COROLLARY 2.16. *The set of triangulations of $C(n+2d+1, 2d)$ without interior $(d+1)$ -simplices forms a connected subgraph of the flip graph of $C(n+2d+1, 2d)$.*

Proof. Slice mutation involves replacing one vertex of a slice S with another to obtain a new slice S' . Hence, if one considers the triangulations $\pi(S_0)$ and $\pi(S'_0)$, these have all but one d -arc in common. Since two triangulations are related by a bistellar flip if and only if they have all but one d -arc in common by Theorem 1.3, it follows that triangulations related by slice mutation are related by a bistellar flip. Iyama and Oppermann then show that all slices are connected by slice mutation [26, Theorem 5.27]. Hence this implies that the class of triangulations without interior $(d+1)$ -simplices is connected by bistellar flips, by Theorem 2.8. $\square$

REMARK 2.17. Rambau proves in [32] that the set of triangulations of any cyclic polytope is connected via bistellar flips. Corollary 2.16 does not follow from this, since it concerns the subclass of triangulations of $C(n+2d+1, 2d)$ which do not have interior $(d+1)$ -simplices. That is, Corollary 2.16 gives us that two triangulations $\mathcal{T}$ and $\mathcal{T}'$ of $C(n+2d+1, 2d)$ without interior $(d+1)$ -simplices are connected via bistellar flips through triangulations which also have no interior $(d+1)$ -simplices. On the other hand, the result from [32] only gives us that $\mathcal{T}$ and $\mathcal{T}'$ are connected via bistellar flips through arbitrary triangulations, which may possibly have interior $(d+1)$ -simplices.

3. A COMBINATORIAL CRITERION FOR MUTATION

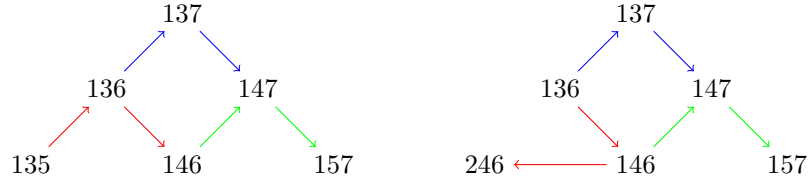
Given a triangulation $\mathcal{T}$ of $C(n+2d+1, 2d)$, we say that a d -arc A of $\mathcal{T}$ is *mutable* if there is a bistellar flip $\mathcal{T}'$ of $\mathcal{T}$ such that $\text{d-arcs}(\mathcal{T}) \setminus \{A\} = \text{d-arcs}(\mathcal{T}') \setminus \{B\}$. It is clear that here A and B must intertwine. For $d = 1$, where triangulations of $C(n+2d+1, 2d)$ are triangulations of convex $(n+3)$ -gons, all d -arcs are mutable. But this is not true for $d > 1$. In this section, we prove a criterion for identifying the mutable d -arcs of a triangulation $\mathcal{T}$ from its quiver $Q(\mathcal{T})$. This then leads us to a rule for mutating cut quivers at vertices which are neither sinks nor sources. From the perspective of polyhedral geometry, we again see how properties of the triangulation are encoded in the quiver; from the algebraic perspective, we see what a theory of higher-dimensional quiver mutation might look like.

We begin with some motivating observations concerning cuts. We explain how a cut quiver may be decomposed into distinguished cut cycles, and observe that an arc of the triangulation is mutable if and only if it does not occur in the middle of a distinguished cut cycle.

LEMMA 3.1. *The arrows of each $(d+1)$ -cycle in $Q^{(d,n)}$ must be labelled exactly once by each element of $\{F_0, F_1, \dots, F_d\}$.*

Proof. To see this, consider a $(d+1)$ -cycle with labels $F_{i_0}, F_{i_1}, \dots, F_{i_d}$. Then we must have $\sum_{j=0}^d F_{i_j} = 0$, since the cycle finishes where it started. Hence $\{F_{i_0}, F_{i_1}, \dots, F_{i_d}\} = \{F_0, F_1, \dots, F_d\}$. $\square$

FIGURE 8. Mutability via distinguished cut $(d+1)$ -cycles



We call a $(d+1)$ -cycle of $Q^{(d,n)}$ *distinguished* if the arrows are labelled in the cyclic order $F_d, F_{d-1}, \dots, F_0$. Given a cut C , the distinguished cut $(d+1)$ -cycles of Q_C are the paths that result from removing the arrows of C from the distinguished cycles of $Q^{(d,n)}$. The following observation is entailed by the main result of the subsequent section.

OBSERVATION 3.2. *Given a triangulation $\mathcal{T}$ whose quiver $Q(\mathcal{T})$ is a cut of $Q^{(d,n)}$, the mutable d -arcs of $\mathcal{T}$ are precisely the d -arcs which do not lie in the middle of a distinguished cut $(d+1)$ -cycle.*

EXAMPLE 3.3. The reader can check that in the left-hand triangulation in Figure 8 the mutable 2-arcs are 135, 146, and 157, whilst in the right-hand triangulation the mutable 2-arcs are 246, 136, and 157. In these figures we draw each distinguished cut 3-cycle in a different colour.

3.1. GENERAL TRIANGULATIONS. Cut quivers have a very particular form and it is this that allows us to determine the distinguished cut $(d+1)$ -cycles of the quiver, and then to use these to determine the mutable d -arcs of the triangulation. In general, quivers of triangulations may be much more complicated than cut quivers. Nevertheless, we may generalise Observation 3.2 to arbitrary triangulations of even-dimensional cyclic polytopes using the following notion.

DEFINITION 3.4. *Given a triangulation $\mathcal{T}$ of $C(n+2d+1, 2d)$ and fixing a cyclically shifted order of $[n+2d+1]$, a path*

$$(\dots, a_{i-1}, a_i, a_{i+1}, \dots) \rightarrow (\dots, a_{i-1}, b_i, a_{i+1}, \dots) \rightarrow (\dots, a_{i-2}, b_{i-1}, b_i, a_{i+1}, \dots)$$

in $Q(\mathcal{T})$ is retrograde at $(a_0, \dots, a_{i-1}, b_i, a_{i+1}, \dots, a_d)$ if and only if $a_{i-1} < b_{i-1} < a_i$.

A path

$$A_0 \rightarrow A_1 \rightarrow \dots \rightarrow A_l \rightarrow A_{l+1}$$

is retrograde if $A_{i-1} \rightarrow A_i \rightarrow A_{i+1}$ is retrograde at A_i for all $i \in [l]$. For $i \in [l]$, we say that A_i is in the middle of this retrograde path. We consider paths consisting of a single arrow to be trivially retrograde. We say that a retrograde path is maximal if it is not contained in any longer retrograde paths.

REMARK 3.5. The distinguished cut $(d+1)$ -cycles in a cut quiver are maximal retrograde paths. Arrows in this cut cycle are labelled $F_{l+1}, \dots, F_d, F_0, \dots, F_{l-1}$ for some l . The arrows in $Q(\mathcal{T})$ labelled by F_i are of the form $A \rightarrow A + E_i$, and we always have that $a_i + 1 < a_{i+1}$. Hence, we obtain that these paths are retrograde. These paths are, furthermore, maximally retrograde, since the next arrow in the path at either end would have to be labelled by F_l , but this is precisely the arrow that has been cut out.

LEMMA 3.6. *Every arrow in $Q(\mathcal{T})$ is contained in a unique maximal retrograde path.*

Proof. If an arrow follows $A \rightarrow A + rE_i$, then must be of the form $A + rE_i \rightarrow A + rE_i + sE_j$ with $j = i - 1$. If an arrow precedes it, then it must be of the form $A - sE_j \rightarrow A$ with $j = i + 1$. Both such arrows must be unique, by Corollary 2.4. Hence, there is only one way to extend an arrow to a maximal retrograde path. $\square$

PROPOSITION 3.7. *The maximal length of a retrograde path in $Q(\mathcal{T})$ is d .*

Proof. Suppose for contradiction that we have a retrograde path in $Q(\mathcal{T})$ of length $(d + 1)$. By re-ordering, we can represent this in the form

$$(a_0, a_1, \dots, a_d) \rightarrow (a_0, a_1, \dots, a_{d-1}, b_d) \rightarrow \dots \rightarrow (a_0, b_1, b_2, \dots, b_d) \rightarrow (b_0, \dots, b_d).$$

Since this path is retrograde, we have $a_i < b_i < a_{i+1}$ for all $i \in [d]$. But this implies that $(a_0, a_1, \dots, a_d)$ and $(b_0, b_1, \dots, b_d)$ are intertwining. $\square$

A d -arc A is mutable precisely if there exists a d -arc B which intertwines with it but which does not intertwine with any other d -arc in the triangulation. If such a d -arc B does not exist, then A is not mutable. Hence, we consider the collection of d -arcs of a triangulation which intertwine with a given d -arc outside the triangulation.

LEMMA 3.8. *Let $B \in \circlearrowleft \mathbf{I}_{n+2d+1}^d \setminus \mathbf{d}\text{-arcs}(\mathcal{T})$. Let $Q_B(\mathcal{T})$ be the full subquiver of $Q(\mathcal{T})$ with vertex set*

$$Q_B(\mathcal{T})_0 = \{A \in \mathbf{d}\text{-arcs}(\mathcal{T}) \mid A \bowtie B\}.$$

Then $Q_B(\mathcal{T})$ is connected.

Proof. Let $\mathcal{T}_B$ be the collection of $2d$ -simplices of $\mathcal{T}$ which have a d -face intertwining with the d -arc B . Let $T \in \mathcal{T}_B$. We first show that the set of d -faces of T is connected in $Q_B(\mathcal{T})$. Hence, let A, A' be two d -faces of T which intertwine with B . Then A and A' must have a common vertex, since they are both faces of the same $2d$ -simplex, so, by re-ordering, we can assume $a_0 = a'_0$. We know that A and B must be intertwining, so we may also assume that

$$a_0 < b_0 < a_1 < b_1 < \dots < b_{d-1} < a_d < b_d.$$

Since $a_0 = a'_0$ and A' also intersects B , we also have that

$$a'_0 < b_0 < a'_1 < b_1 < \dots < b_{d-1} < a'_d < b_d.$$

Let $c_i = \min\{a'_i, a_i\}$, where this minimum is taken in the linearly ordered set between b_{i-1} and b_i . Then $C \in \mathcal{T}$ since it is a d -face of T . Moreover, it is in $Q_B(\mathcal{T})$. There is a path $C \rightsquigarrow A$ in $Q_B(\mathcal{T})$ due to Definition 2.1 since, by construction, $(C - \mathbf{1}) \bowtie A$. There is likewise a path $C \rightsquigarrow A'$ in $Q_B(\mathcal{T})$. Therefore, A and A' are connected to each other in $Q_B(\mathcal{T})$. Hence, any two d -arcs lying in a common $2d$ -simplex are connected by a walk in $Q_B(\mathcal{T})$.

We now show that the d -arcs in $Q_B(\mathcal{T})$ which lie in different $2d$ -simplices are connected with each other. Let $T, T' \in \mathcal{T}_B$. Recall from Section 1.2 that we use $|\cdot|$ to denote geometric realisation of simplices, so that $|B|$ is the geometric realisation of B . If one chooses points $x \in |T| \cap |B|$ and $x' \in |T'| \cap |B|$, then the line segment $\overline{xx'}$ connecting x and x' must lie entirely within $|B|$, since $|B|$ is convex. If one travels from $|T|$ to $|T'|$ along $\overline{xx'}$, then one runs through a series of $2d$ -simplices $|T| = |T_0|, |T_1|, \dots, |T_r| = |T'|$ where each pair of $2d$ -simplices $|T_{l-1}|$ and $|T_l|$ shares a common face $|U_l|$ which must also intersect $|B|$. Then, by the description of the circuits of $C(n + 2d + 1, 2d)$ from Section 1.2 as pairs of intertwining d -arcs, there must be a d -arc J_l within U_l such that $B \bowtie J_l$. Let $Q_B(\mathcal{T})|_{T_{l-1}}$ and $Q_B(\mathcal{T})|_{T_l}$ respectively denote the full subquivers of $Q_B(\mathcal{T})$ consisting of the d -arcs lying in T_{l-1} and T_l . We therefore have that $Q_B(\mathcal{T})|_{T_{l-1}}$ and $Q_B(\mathcal{T})|_{T_l}$ share the d -arc J_l . Moreover, this means that $Q_B(\mathcal{T})|_T$ and $Q_B(\mathcal{T})|_{T'}$ are connected to each other in $Q_B(\mathcal{T})$. Hence $Q_B(\mathcal{T})$ itself is connected. $\square$

REMARK 3.9. Lemma 3.8 may also be seen quickly using an algebraic argument. [30, Theorem 5.6] implies that $Q_B(\mathcal{T})$ must be the support of an indecomposable module, and so must be connected.

This gives the following useful corollary, which implies that in order to check whether one can mutate a d -arc A to a d -arc B , it suffices only to check whether the d -arcs adjacent to A in the quiver intertwine with B , rather than checking all d -arcs for whether they intertwine with B .

COROLLARY 3.10. *Let $A \in \mathcal{T}$ and $B \in \circlearrowleft \mathbf{I}_{n+2d+1}^d$ with $A \delta B$. If there is an $A' \in \mathbf{d}\text{-arcs}(\mathcal{T})$ with $A' \neq A$ and $A' \delta B$, then there is an $A'' \in \mathbf{d}\text{-arcs}(\mathcal{T})$ with $A'' \delta B$, such that A and A'' are adjacent in $Q(\mathcal{T})$.*

Proof. Suppose that we are in the situation described. We know from Lemma 3.8 that $Q_B(\mathcal{T})$ is connected, and the set-up gives us that it contains at least two vertices, one of which is A . Hence there is a vertex of $Q_B(\mathcal{T})$ which is adjacent to A . $\square$

We can now prove the main theorem of this section.

THEOREM 3.11. *Let $\mathcal{T}$ be a triangulation of $C(n+2d+1, 2d)$. Then a d -arc of $\mathcal{T}$ is mutable if and only if it is not in the middle of a maximal retrograde path in $Q(\mathcal{T})$.*

Proof. Let $A \in \mathbf{d}\text{-arcs}(\mathcal{T})$. For each entry a_i in A , let $Z_i = (a_0, a_1, \dots, a_{i-1}, z_i, a_{i+1}, a_{i+2}, \dots, a_d)$ be the d -arc of this form in $\mathcal{T}$ such that there is an arrow $Z_i \rightarrow A$, if there is such a d -arc. Similarly, let $B_i = (a_0, a_1, \dots, a_{i-1}, b_i, a_{i+1}, a_{i+2}, \dots, a_d)$ be the d -arc of this form in $\mathcal{T}$ such that there is an arrow $A \rightarrow B_i$, if there is such a d -arc.

The d -arc A is mutable if and only if there exists $C \in \circlearrowleft \mathbf{I}_{n+2d+1}^d$ such that $A \delta C$ but such that there is no $A' \in \mathcal{T}$ with $A' \neq A$ with $A' \delta C$. By Corollary 3.10, it is necessary and sufficient that we do not have $Z_i \delta C$ or $B_i \delta C$ for any $i \in \{0, 1, \dots, d\}$. If $A \delta C$, then, since

$$a_0 < c_0 < a_1 < c_1 < \dots < a_d < c_d,$$

we have that Z_{i+1} and C do not intertwine if and only if $z_{i+1} \leq c_i$. Similarly, B_i and C do not intertwine if and only if $c_i \leq b_i$.

Let $z_{i+1} = a_i + 1$ if Z_{i+1} does not exist and let $b_i = a_{i+1} - 1$ if B_i does not exist. The rationale for this is that these are respectively the minimal and maximal values for c_i . Then, by the above reasoning, A is mutable precisely if there exists

$$C \in [z_1, b_0] \times [z_2, b_1] \times \dots \times [z_d, b_{d-1}] \times [z_0, b_d].$$

So A is mutable if and only if this product is non-empty. But the product is non-empty if and only if $z_{i+1} \leq b_i$ for all i , which is precisely the condition that none of the paths $Z_{i+1} \rightarrow A \rightarrow B_i$ are retrograde. $\square$

COROLLARY 3.12. *Let $\mathcal{T}$ be a triangulation of $C(n+2d+1, 2d)$ and $A \in \mathbf{d}\text{-arcs}(\mathcal{T})$ such that A is not in the middle of any maximal retrograde paths. Let Z_i and B_i be as in the proof of Theorem 3.11. Then $z_{i+1} = b_i$ for all i and $(b_0, b_1, \dots, b_d)$ replaces A in the bistellar flip at A .*

Proof. We know from the proof of Theorem 3.11 that any element of

$$[z_1, b_0] \times [z_2, b_1] \times \dots \times [z_d, b_{d-1}] \times [z_0, b_d]$$

may replace A in a bistellar flip. But, we have that the d -arc which can replace A in a bistellar flip must be unique. Hence $z_{i+1} = b_i$ for all i and the unique element of the product must replace A in the bistellar flip. $\square$

FIGURE 9. Triangulation of $C(8, 4)$

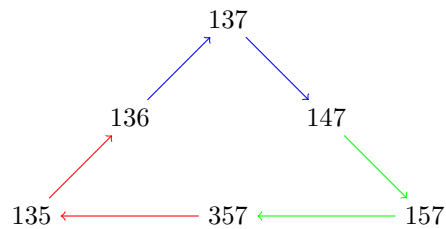
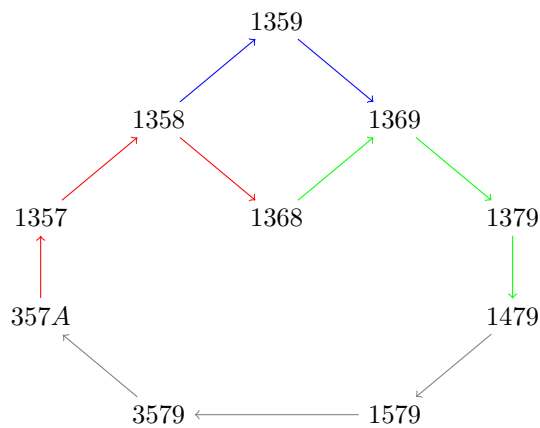


FIGURE 10. Triangulation of $C(10, 6)$



Theorem 3.11 and Corollary 3.12 make it much easier to compute bistellar flips of triangulations of even-dimensional cyclic polytopes by hand. This also makes it easier to compute mutations of cluster-tilting objects of A_n^d by hand.

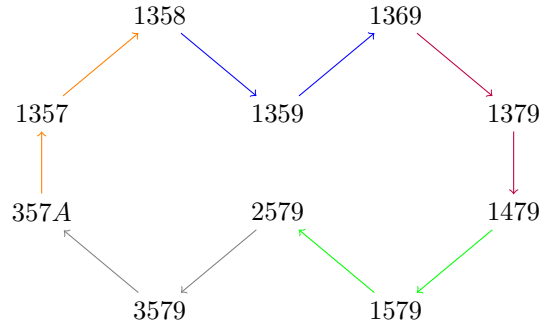
EXAMPLE 3.13. We provide examples of how one may use this criterion to identify the mutable d -arcs of a triangulation. We represent maximal retrograde paths using consecutive arrows of the same colour.

Considering the triangulation of $C(8, 4)$ given in Figure 9, the mutable 2-arcs are 136, 147, and 357.

Note that maximal retrograde paths are not always of length d . This is shown by the triangulation of $C(10, 6)$ given in Figure 10. (We use ‘A’ to denote 10.) The mutable d -arcs of this triangulation are 357A, 1368, 1479.

One can also illustrate Corollary 3.12. Consider the d -arc 1368 in Figure 10. This is mutable by Theorem 3.11, so we can compute what d -arc it is exchanged for. Between 1 and 3 we must have 2 and between 6 and 8 we must have 7. Then, between 3 and 6 we must have 5 since 1368 is adjacent to 1358. Similarly, between 8 and 1 we must have 9, because 1368 is adjacent to 1369. Hence performing a bistellar flip at 1368 exchanges this d -arc for 2579. Observe that the retrograde-path analysis makes it easier to compute the bistellar flips of the triangulation. In the resulting triangulation, shown in Figure 11, none of the retrograde paths are of length d ; they are all of length $d - 1$.

FIGURE 11. Triangulation of $C(10, 6)$



3.2. MUTATING CUT QUIVERS. There is a rule for mutating cut quivers at sinks and sources [26], as described in Section 1.3.1. In this section, we extend this rule to allow mutation at vertices which are not in the middle of retrograde paths, but which are not necessarily sinks or sources. In the case where the cut quiver is $Q(\mathcal{T})$ for a triangulation $\mathcal{T}$, we also describe the effect of the mutation on the triangulation $\mathcal{T}$.

For the following lemmas, we let C be a cut of $Q^{(d,n)}$. The purpose of these lemmas is to describe the local structure of a cut quiver around a vertex which is not in the middle of a distinguished cut $(d+1)$ -cycle. We then use our knowledge of this local structure to describe the effect of mutation at that vertex. Recall the notion of types of arrows from Section 1.3.

LEMMA 3.14. *If a vertex x of $Q_C^{(d,n)}$ is the source of one distinguished cut $(d+1)$ -cycle and the sink of another distinguished cut $(d+1)$ -cycle, then the arrows cut out of the two cycles are of the same type.*

Proof. If the arrows cut out are of different types, then they form two consecutive arrows of a $(d+1)$ -cycle, by Lemma 3.1. This $(d+1)$ -cycle therefore has two arrows cut out. But this is a contradiction, since a cut removes precisely one arrow from each $(d+1)$ -cycle. $\square$

LEMMA 3.15. *If a vertex x of $Q_C^{(d,n)}$ is neither a source nor a sink of the quiver, nor in the middle of a distinguished cut $(d+1)$ -cycle, then it is the head of precisely one arrow and the tail of precisely one arrow.*

Proof. Suppose that x is neither a source nor a sink of the quiver, nor in the middle of a distinguished cut $(d+1)$ -cycle. Then, x must be the source of one distinguished cut $(d+1)$ -cycle and the sink of another distinguished cut $(d+1)$ -cycle. Moreover, x is the head of at least one arrow α and the tail of at least one arrow β . Since x is not in the middle of a distinguished cut $(d+1)$ -cycle, the arrows α' and β' succeeding α and preceding β in their respective distinguished $(d+1)$ -cycles must be cut out. By Lemma 3.14, α' and β' have the same type.

Suppose that we have another arrow γ such that x is the head of γ . Then the arrow γ' which succeeds it in the distinguished $(d+1)$ -cycle must be cut out. But then by Lemma 3.14, γ' , α' , and β' all have the same type, so $\gamma = \alpha$. A similar argument can be made for an arrow δ with tail x . $\square$

LEMMA 3.16. *Let $\mathcal{T}$ be a triangulation of $C(n+2d+1, 2d)$ with $A \in \text{d-arcs}(\mathcal{T})$. If $a_i + 2 < a_{i+1}$ for some i , then there is either an arrow*

$$(a_0, \dots, a_i, z_{i+1}, a_{i+2}, \dots, a_d) \rightarrow A$$

or an arrow

$$A \rightarrow (a_0, \dots, a_{i-1}, b_i, a_{i+1}, \dots, a_d).$$

Proof. We use the fact that there is a $2d$ -simplex T of $\mathcal{T}$ such that $e(T) = A$. If we let $T = (a_0, x_1, a_1, x_2, \dots, a_{d-1}, x_d, a_d)$, then we must have either $x_{i+1} > a_i + 1$ or $x_{i+1} < a_{i+1} - 1$, since $a_i + 2 < a_{i+1}$. In the former case, we must have a path $(a_0, a_1, \dots, a_i, x_{i+1}, a_{i+2}, a_{i+3}, \dots, a_d) \rightsquigarrow A$ comprised of arrows of type $i + 1$; in the latter case, we must have a path $A \rightsquigarrow (a_0, a_1, \dots, a_{i-1}, x_i, a_{i+1}, a_{i+2}, \dots, a_d)$ of type i . This establishes the claim. $\square$

LEMMA 3.17. *If a vertex A is the head of precisely one arrow α_d and the tail of precisely one arrow β_1 , and not in the middle of a distinguished cut $(d + 1)$ -cycle, then there is an a_i such that for $j \notin \{i, i - 1\}$, $a_j + 2 = a_{j+1}$.*

Proof. The arrow α_d must be the final arrow in a distinguished cut $(d + 1)$ -cycle $\alpha_1 \alpha_2 \dots \alpha_d$. The arrow β_1 must be the first arrow in a distinguished cut $(d + 1)$ -cycle $\beta_1 \beta_2 \dots \beta_d$. Let α_0 and β_0 be the respective arrows cut out of these $(d + 1)$ -cycles.

We know from Lemma 3.14 that α_0 and β_0 are of the same type. We let this type be i . Then β_1 has type $i - 1$ and α_d has type $i + 1$. Suppose that there is a $j \notin \{i, i - 1\}$ such that $a_j + 2 < a_{j+1}$. Then, by Lemma 3.16, we have that there is either an arrow

$$(a_0, \dots, a_j, z_{j+1}, a_{j+2}, \dots, a_d) \rightarrow A$$

or an arrow

$$A \rightarrow (a_0, \dots, a_{j-1}, b_j, a_{j+1}, \dots, a_d).$$

We then must have that $j + 1 = i + 1$ or $j = i - 1$, respectively, but both are contradictions. $\square$

With these lemmas in place, we can now describe the effect of mutation on a cut quiver at a vertex which is not in the middle of a distinguished cut $(d + 1)$ -cycle, but is not a sink or a source. Since the description of mutation at sinks and sources is covered in [26], as described in Section 1.3.1, and we know that mutation at vertices in the middle of distinguished cut $(d + 1)$ -cycles is not possible by Theorem 3.11, this completes the description of mutation of cut quivers.

PROPOSITION 3.18. *Let $\mathcal{T}$ be a triangulation of $C(n + 2d + 1, 2d)$ such that $Q(\mathcal{T})$ is isomorphic to $Q_C^{(d,n)}$ for a cut C . Let A be a vertex of $Q(\mathcal{T})$ which is neither a sink nor a source, but is still not in the middle of any retrograde paths. Let $\mathcal{T}'$ be the result of performing a bistellar flip at A in $\mathcal{T}$. Then we have the following.*

- (1) *In the bistellar flip A is replaced by $(a_0 + 1, a_1 + 1, \dots, a_{i-1} + 1, a_{i+1} - 1, a_{i+1} + 1, \dots, a_d + 1)$, where i is the type of the arrow cut out of the distinguished cut $(d + 1)$ -cycles at A .*
- (2) *$Q(\mathcal{T}')$ is obtained from $\mathcal{T}$ by removing from C the arrows beginning or ending at A and adding the arrows of $Q_1(\mathcal{T}) \setminus C$ which begin or end at A .*

Proof. By Lemma 3.15 we have that A is the source of precisely one arrow and the sink of precisely one arrow. Moreover, by Lemma 3.17, the vertex A is such that there is an a_i such that for $j \notin \{i, i - 1\}$, we have $a_j + 2 = a_{j+1}$. We assume, by re-ordering, that $i = 0$. Furthermore, the distinguished cut $(d + 1)$ -cycle beginning at A must look like

$$(a_0, a_1, \dots, a_d) \rightarrow (a_0, a_1, \dots, a_{d-1}, a_d + 1) \rightarrow \dots \rightarrow (a_0, a_1 + 1, a_2 + 1, \dots, a_d + 1),$$

and the distinguished cut $(d + 1)$ -cycle ending at A must look like

$$(a_0, a_1 - 1, a_2 - 1, \dots, a_d - 1) \rightarrow \dots \rightarrow (a_0, a_1 - 1, a_2, a_3, \dots, a_d) \rightarrow (a_0, a_1, \dots, a_d),$$

by Lemma 2.9. It follows from Corollary 3.12 that A mutates to

$$(a_1 - 1, a_2 - 1, \dots, a_d - 1, a_d + 1) = (a_1 - 1, a_1 + 1, \dots, a_{d-1} + 1, a_d + 1),$$

settling (1).

Then we have arrows in $Q(\mathcal{T}')$ given by

$$(a_1 - 1, a_1 + 1, a_2 + 1, \dots, a_d + 1) \leftarrow (a_0, a_1 + 1, a_2 + 1, \dots, a_d + 1),$$

and

$$(a_0, a_1 - 1, a_2 - 1, \dots, a_d - 1) \leftarrow (a_d + 1, a_1 - 1, a_2 - 1, \dots, a_d - 1),$$

since, by assumption, there are no arrows of type 0 ending at $(a_0, a_1 + 1, \dots, a_d + 1)$ or $(a_d + 1, a_1 - 1, a_2 - 1, \dots, a_d - 1)$. Indeed, these new arrows are precisely the arrows of C which begin or end at A . On the other hand, the arrows in $Q_1(\mathcal{T}) \setminus C$ beginning or ending at A are absent from $Q_1(\mathcal{T}')$. Since $a_j + 2 = a_{j+1}$ for $j \notin \{i, i - 1\}$, there can be no other new arrows in $Q(\mathcal{T}')$, thus settling (2). $\square$

EXAMPLE 3.19. Proposition 3.18 can be verified by mutating the left-hand triangulation in Figure 8 to obtain the triangulation in Figure 9.

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